Quantum Gravitational Dynamics

Chapter 4: Gravitational Energy

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Abstract

General Relativity possesses no well-defined local energy-momentum tensor for the gravitational field. QGD resolves this century-old problem: the graviton field σ_μ admits a true Noether tensor $T^{\mu\nu}_{\text{QGD}} = \nabla^\mu\sigma_\alpha\nabla^\nu\sigma^\alpha - \frac{1}{2}g^{\mu\nu}(\nabla\sigma)^2 + \ell_Q^2$-corrections, which is coordinate-independent, positive-definite, and locally conserved. We derive the Schwarzschild energy profile, two-body binding energy, exact gravitational wave energy, and the gravitational Poynting vector.

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1 The Problem in General Relativity

The gravitational field in GR carries energy — gravitational waves heat up detectors — yet the theory provides no local, covariant energy-momentum tensor for this energy. The standard remedies are:

- The Einstein pseudotensor $t_{\mu\nu}$: coordinate-dependent, can be set to zero at any point by choosing free-fall coordinates.

- The Landau–Lifshitz pseudotensor: symmetric but still non-covariant.
- The Bondi mass: defined only at null infinity; cannot locate energy in space.
- The ADM mass: defined only at spatial infinity.

The root cause is the equivalence principle: in free fall, the metric is locally flat and its first derivatives vanish. Since the pseudotensors are built from ∂g, they vanish too.

QGD resolution: the graviton field σ_μ generates the metric but is not itself a metric component. It cannot be gauged away, just as the Yang–Mills field strength $F_{\mu\nu}$ persists even when the potential A_μ is gauged. This structural shift is what makes localizable energy possible.

2 The QGD Energy-Momentum Tensor

2.1 Derivation from Noether's theorem

The QGD action for the σ-field is

$$S[\sigma] = \int \mathrm{d}^4x \left[\frac{1}{2}\partial_\alpha\sigma_\mu\partial^\alpha\sigma^\mu - V(\sigma) + \frac{\ell_Q^2}{2}(\partial_\alpha\partial_\beta\sigma_\mu)^2 + \mathcal{L}_{\text{matter}}[g[\sigma]] \right] \tag{1}$$

Applying Noether's theorem to translation invariance $x^\mu \to x^\mu + a^\mu$:

$$\boxed{T^{\mu\nu}_{\text{QGD}} = \nabla^\mu\sigma_\alpha\nabla^\nu\sigma^\alpha - \frac{1}{2}g^{\mu\nu}(\nabla\sigma)^2 - g^{\mu\nu}V(\sigma) + \ell_Q^2\text{-corrections}} \tag{2}$$

2.2 Properties

1. **True tensor:** transforms covariantly under all coordinate changes. Cannot be set to zero by any gauge choice.
2. **Conservation:** $\partial_\mu T^{\mu\nu}_{\text{QGD}} = 0$ follows automatically from the field equations. No pseudotensor gymnastics required.
3. **Positive-definite Hamiltonian:**

$$H[\sigma,\pi] = \int \mathrm{d}^3x \left[\frac{1}{2}\pi_\mu\pi^\mu + \frac{1}{2}(\nabla\sigma_\mu)^2 + V(\sigma) + \frac{\ell_Q^2}{2}(\nabla^2\sigma_\mu)^2 \right] \geq 0 \tag{3}$$

   Equality holds if and only if $\sigma_\mu = 0$ (flat Minkowski). This is manifest; it supersedes the 64-year effort culminating in the Schoen–Yau proof (1979).

3 Local Energy Density

The T^{00} component gives the gravitational energy density at every spacetime point:

$$\boxed{\rho_{\text{grav}}(\mathbf{x}) = \underbrace{\frac{1}{2}\dot\sigma_\mu\dot\sigma^\mu}_{\text{kinetic}} + \underbrace{\frac{1}{2}(\nabla\sigma_\mu)^2}_{\text{gradient}} + \underbrace{V(\sigma)}_{\text{potential}} + \underbrace{\frac{\ell_Q^2}{2}(\nabla^2\sigma_\mu)^2}_{\text{quantum stiffness}}} \tag{4}$$

Each term is manifestly non-negative (for $V \geq 0$). This is a well-defined scalar field at every point in spacetime.

4 Energy Density of the Schwarzschild Field

For a Schwarzschild black hole of mass M:

$$\sigma_t(r) = \sqrt{\frac{r_s}{r}}, \qquad r_s = \frac{2GM}{c^2} \tag{5}$$

The gradient energy density is:

$$\rho_{\text{grav}}(r) = \frac{1}{2}\left(\frac{\mathrm{d}\sigma_t}{\mathrm{d}r}\right)^2 = \frac{1}{2} \cdot \frac{r_s}{4r^3} = \boxed{\frac{GM}{4c^2r^3}} \tag{6}$$

Physical interpretation:

- Energy concentrated near the horizon: $\rho \propto r^{-3}$.
- At $r = r_s$: $\rho = c^4/(32G^2M^2)$.
- At $r = 10\,r_s$: three orders of magnitude smaller.
- No energy at the singularity: the quantum stiffness $\ell_Q^2(\nabla^2\sigma)^2$ activates at $r \sim \ell_Q$ and prevents unlimited concentration.

4.1 Newtonian limit

In the weak-field regime with Newtonian potential $\Phi = -GM/r$:

$$\rho_{\text{grav}} = \frac{1}{2}(\nabla\sigma_t)^2 = \frac{(\nabla\Phi)^2}{2c^2} \propto \frac{g^2}{8\pi G} \tag{7}$$

This is exactly the classical gravitational field energy density, derived here from the σ-field structure.

5 Two-Body Binding Energy

For two masses M_1, M_2 separated by distance d, the σ-field superposition is:

$$\sigma_t(\mathbf{x}) = \sqrt{\frac{2GM_1}{c^2|\mathbf{x}-\mathbf{x}_1|}} + \sqrt{\frac{2GM_2}{c^2|\mathbf{x}-\mathbf{x}_2|}} \tag{8}$$

The total field energy splits as $E = E_1 + E_2 + E_{\text{cross}}$, where the cross term from $\int \nabla\sigma_1 \cdot \nabla\sigma_2 \,\mathrm{d}^3x$ gives:

$$\boxed{E_{\text{binding}} = -\frac{GM_1M_2}{d}} \tag{9}$$

Newton's gravitational binding energy emerges automatically from the field-gradient structure.

6 Gravitational Wave Energy

6.1 Exact, point-wise definition

A gravitational wave is a radiation mode $\sigma_\mu^{\text{GW}} = \epsilon_\mu e^{ik\cdot x}$ with $k^2 = 0$. The energy density is:

$$\boxed{\rho_{\text{GW}} = \frac{1}{2}(\partial\sigma^{\text{GW}})^2 = \frac{1}{2}\omega^2|\epsilon|^2} \tag{10}$$

This is exact, point-wise defined, and gauge-invariant. No averaging over wavelengths required.

6.2 Recovery of Isaacson

In the weak-field limit, the LIGO observable gives $h_{ij}^{\mathrm{TT}} \sim -2\sigma_i\sigma_j$, so:

$$\rho_{\mathrm{GW}} \longrightarrow \frac{c^2}{32\pi G}\langle \dot{h}^2 \rangle \tag{11}$$

recovering the Isaacson (1968) formula, which in GR requires an averaging procedure that is approximate and valid only in the short-wave limit.

6.3 The gravitational Poynting vector

Energy flux at any radius (not just infinity):

$$\boxed{\mathbf{S}_{\mathrm{grav}} = \dot{\sigma}_\mu \nabla \sigma^\mu, \qquad \frac{\mathrm{d}E}{\mathrm{d}t} = \oint_S \mathbf{S}_{\mathrm{grav}} \cdot \mathrm{d}\mathbf{A}} \tag{12}$$

This enables energy-flow tracking from the near-zone orbital region through the wave zone to infinity — a capability absent in GR.

7 The Q_μ Self-Energy Term

The master field equation $\Box_g \sigma_\mu = Q_\mu + G_\mu + T_\mu + \kappa \ell_Q^2 \Box_g^2 \sigma_\mu$ contains the self-interaction source:

$$Q_\mu = \sigma_\mu(\nabla_\alpha \sigma_\beta \nabla^\alpha \sigma^\beta) + (\nabla_\mu \sigma_\alpha)(\sigma_\beta \nabla^\beta \sigma^\alpha) \tag{13}$$

Q_μ is the gravitational field's self-energy current: the contribution from the field's own energy density. The pattern across gauge theories:

Theory	Self-coupling	Consequence
Maxwell ($Q = 0$)	None	Linear, no self-energy
Yang–Mills ($Q \sim A \cdot F$)	Gluon self-interaction	Confinement
QGD ($Q \sim \sigma(\partial\sigma)^2$)	Graviton self-energy	Black holes, GW energy

8 Comparison: GR vs. QGD Energy

Property	GR	QGD
Local energy density	Pseudotensor only	Scalar $\rho = \frac{1}{2}(\partial\sigma)^2$
True tensor	Does not exist	Noether $T^{\mu\nu}_{\mathrm{QGD}}$
Conservation	Coordinate-dependent	Automatic
Positive energy	Schoen–Yau (1979, at ∞)	Manifest: $H \geq 0$ locally
GW energy	Isaacson averaging	Exact: $\rho_{\mathrm{GW}} = \frac{1}{2}\omega^2 \lvert\epsilon\rvert^2$
Black hole energy	ADM mass (at ∞)	Distributed: $\rho \propto r^{-3}$
Binding energy	Linearised GR; source undefined	$-GM_1M_2/r$ from gradients
Energy flux	Only at null infinity	Poynting vector at any r

9 Conclusions

The century-long mystery of gravitational energy was a consequence of using the metric as the fundamental variable. In QGD:

1. A true energy-momentum tensor $T^{\mu\nu}_{\text{QGD}}$ exists, derived by Noether's theorem.
2. Energy density $\rho_{\text{grav}} = \frac{1}{2}(\partial\sigma)^2$ is a well-defined scalar field at every point.
3. Conservation is automatic; positive energy is manifest.
4. Gravitational wave energy is exact and point-wise.
5. Two-body binding energy $-GM_1M_2/r$ emerges from field-gradient cross terms.
6. The gravitational Poynting vector tracks energy flow at any radius.